MAHATMA GANDHI UNIVERSITY, KOTTAYAMMGU-UGP (HONOURS)

REGULAR /IMPROVEMENT/ REAPPEARANCEEXAMINATION

NOVEMBER2025

FIRST SEMESTER

Discipline Specific Core Course (DSC) - MG1DSCSAC100 - INTRODUCTIONTO STRENGTH AND CONDITIONING

(2024 ADMISSION ONWARDS)

PART A

1. a) Power

2. b) Kettlebells, medicine balls, and elastic bands

3. b) Performing leg swings and arm circles

4. b) Understanding and applying strategies during competition

5. c) A Medical Professional

6. b) Height and weight

7. b) Bradycardia

8. c) Improve muscle strength while protecting healing tissues

9. b) To improve performance and increase physical fitness

10. b) Task-oriented practice

PART B

11. Strength training/resistance training/endurance/flexibility/speed & agility (any one).

12. Effective communication among strength coaches, head coaches, physiotherapists, and nutritionists ensures that all training, recovery, medical, and nutritional strategies align with the athlete's needs. Collaboration prevents conflicting workloads, supports injury prevention, accelerates return-to-play decisions, and optimizes overall performance by creating a unified, athlete-centered plan.

13. Injury prevention supports continual skill development by keeping athletes healthy across stages. It reduces time lost to injuries, promotes proper movement patterns early, and helps ensure athletes can progress safely and consistently through the LTAD phases.

14. A strength and conditioning coach designs and implements training programs to improve athletes' strength, power, speed, endurance, flexibility, and overall performance while also focusing on injury prevention and safe, progressive physical development.

15.Yes. Strength and conditioning programs improve cardiovascular function by enhancing circulation, reducing blood pressure, improving heart efficiency, and supporting overall metabolic health. Combined resistance and aerobic training is especially beneficial for older adults.

16. Insufficient recovery can lead to **fatigue, performance decline, overreaching or overtraining**, increased stress hormone levels, weakened immunity, and higher risk of injury.

17. BMI, skinfolds, BIA, DEXA, underwater weighing, waist-hip ratio.

18.

- **Flexibility**: The ability of a muscle or muscle group to lengthen.
- **Mobility**: The ability of a joint to move actively through its full range of motion.
- **Importance**: Flexibility helps prevent tightness and reduces injury risk, while mobility enables efficient, powerful, and technically sound movement, improving performance.

19. Apply **Specific Adaptation to Imposed Demands** by designing training that mimics the sport's metabolic and mechanical requirements. For example:

- High-volume, moderate-intensity aerobic runs
- Tempo runs and intervals to improve lactate threshold
- Running-specific strength training (e.g., posterior chain, core)
- Technique work focusing on running economy
- Gradual progression of mileage to meet endurance demands
  The program should develop exactly the adaptations required for endurance running

20. Conditioning refers to the development of energy systems (aerobic and anaerobic) to meet sport-specific demands. Proper conditioning improves stamina, speed, recovery between efforts, and overall performance while reducing fatigue-related injury risk.

21. A major trend was the integration of scientific research and evidence-based training, including the rise of periodization, biomechanical analysis, and the use of sport science technology to optimize training.

22. Muscular imbalance occurs when opposing muscle groups differ in strength, flexibility, or activation. This can alter joint mechanics, increase stress on ligaments and tendons, and lead to overuse injuries or acute strains due to uneven force distribution.

PART C

23. Skill-related physical fitness includes six components:

- **Agility** – The ability to rapidly change direction with control.
  Relevance: Important for sports requiring quick cuts, dodging, and rapid position changes (e.g., basketball, soccer).
- **Balance** – The ability to maintain stability, whether static or dynamic.
  Relevance: Crucial for gymnastics, skiing, martial arts, and reducing injury risk.
- **Coordination** – The ability to use body parts together smoothly and efficiently.
  Relevance: Essential for skills like hitting, catching, dribbling, or complex movement patterns.
- **Power** – The ability to exert force quickly (strength × speed).
  Relevance: Critical for sprinting, jumping, throwing, and explosive movements.
- **Reaction Time** – How quickly an individual responds to a stimulus.
  Relevance: Important for starting sprints, responding to opponents, or reacting to game situations.
- **Speed** – The ability to move quickly from one point to another.
  Relevance: Vital for sprinting, breakaways, chasing plays, and overall athletic effectiveness.

24. Common methods for assessing body composition include:

1) Skinfold measurements (using calipers)
2) Bioelectrical Impedance Analysis (BIA)
3) Dual-Energy X-ray Absorptiometry (DEXA)
4) Hydrostatic (underwater) weighing
5) Air displacement plethysmography (e.g., BodPod)
6) Ultrasound body composition analysis
7) BMI (basic but does *not* truly measure body composition—only used as a general indicator)

25. Health-related physical fitness consists of **five components**:

1) **Cardiorespiratory Endurance** – The ability of the heart, lungs, and circulatory system to supply oxygen during prolonged activity.

Importance: Reduces the risk of heart disease, improves energy levels, and enhances overall health.

2) **Muscular Strength** – The maximum force a muscle can exert.
   Importance: Supports daily activities, prevents injury, and maintains functional independence.
3) **Muscular Endurance** – The ability of a muscle to sustain repeated contractions over time.
   Importance: Helps with posture, reduces fatigue, and supports long-duration tasks.
4) **Flexibility** – The range of motion at a joint.
   Importance: Prevents stiffness, decreases injury risk, and improves mobility and posture.
5) **Body Composition** – The ratio of fat mass to lean mass.
   Importance: Healthy levels reduce the risk of chronic diseases such as diabetes, hypertension, and obesity.

26. Programs must be **individualized** and consider the following:

- **Health and Medical Status**

  Obtain medical clearance.

  Understand limitations, medications, and contraindications.

- **Progressive and Low-Impact Training**

  Start with low intensity and gradually progress.

  Include controlled resistance exercises to avoid excessive strain.

- **Functional Movements**

  Focus on balance, mobility, and daily living activities, especially for older adults and individuals with disabilities.

- **Safety and Supervision**

  Ensure proper technique, appropriate equipment, and safe environments.

  Provide modifications (e.g., seated exercises, assistive devices).

- **Condition-Specific Adjustments**

  For chronic diseases (e.g., diabetes, heart disease), adjust volume and intensity.

  Monitor vital signs when necessary

- **Psychological Considerations**

  Provide motivation, build confidence, and encourage regular participation.

27. **SAID = Specific Adaptation to Imposed Demands**

It states that the body adapts specifically to the type of stress placed on it.

Strength training → increases muscle strength.

Endurance training → improves aerobic capacity.

Sprint training → improves speed and power.

Significance:

- Guides exercise programming to match sport or fitness goals.
- Ensures training is purposeful and efficient.
- Prevents irrelevant or ineffective training
- Helps coaches target specific performance outcomes (e.g., power for sprinters, endurance for long-distance athletes).

28. A proper warm-up helps prevent injuries by:

- **Increasing Blood Flow and Muscle Temperature**

  Warmer muscles contract more efficiently and are less prone to strains or tears.

- **Enhancing Joint Mobility and Flexibility**

  Dynamic movements prepare joints and connective tissues for activity.

- **Improving Nervous System Activation**

  Boosts reaction time, coordination, and muscle readiness.

- **Gradually Increasing Heart Rate**

  Prepares the cardiovascular system for high-intensity work, reducing shock to the body.

- **Mental Preparation**

  Helps athletes focus on upcoming tasks, lowering the risk of careless mistakes that lead to injury.